

Davante

8

ENGLISH

# THE NOTEBOOK IT

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# / 1. Glossary

En el glosario de esta unidad, encontraremos expresiones que nos permitirán utilizar un vocabulario adecuado al contexto en que nos encontramos en nuestra familia profesional. Para ello, estudiaremos los términos que encontramos a continuación.

- **A top-down design** is a method of designing software in which the engineer begins by defining the main user functions and decomposes those functions into subfunctions, until the basic operations of the software are defined.
- **A bottom-up design** is a method of designing software in which the engineer begins with the software's most basic functions and proceeds to more complicated functions, until the higher-order functions of the software have been created.
- **Functional decomposition** is a design philosophy in which a function is decomposed into a number of subfunctions.
- **Stepwise refinement** is a problem-solving approach in software design in which a problem is divided into smaller, more manageable sections.
- **A subfunction** is a component which combines with other subfunctions to make up a function.

# / 2. Listening

En este apartado practicaremos la comprensión oral. En el audio escucharemos una conversación entre un profesor y un alumno sobre distintos términos de diseño. Después, completa las preguntas sobre la conversación.



*Audio Listening. "Design Methods"*

*https://on.soundcloud.com/D3EpLLYyzxT41ar66*



**1. Listen and complete the conversation.**

**Instructor:** Can anyone tell me about the limitations of a bottom-up design? Yes, Linda.

**Student:** A bottom-up design has to be created from scratch. The designer has no previous function to work with.

**Instructor:** Very good. Can you tell me about some of the benefits of a bottom-up design?

**Student:** Um. I'm not sure.

**Instructor:** What about in regards to 1) _____ problems in a design?

**Student:** Oh, well, a bottom-up design is less likely to have problems. But, if it does, it's more difficult to 2) _____ them.

**Instructor:** Great. And what about 3) _____-down designs?

**Student:** Well, top-down designs tend to have more problems. But it's 4) _____ to eliminate them.

**Instructor:** And how would one do that?

**Student:** I'm not sure. Maybe by 5) _____ the software?

**Instructor:** That's correct. It looks like you've been keeping up on the reading.

# / 3. Reading

En este apartado, vamos a trabajar la comprensión escrita con este texto sobre la descripción de un curso de introducción a métodos básicos de diseño de software. Lee el texto, señala el nuevo vocabulario y realiza el ejercicio que se muestra a continuación.

### CMSC 1500
### INTRODUCTION TO DESIGN METHODS

This course is an introduction to basic software design methods. It will teach students about **design methods** in both **idealistic** and realistic settings. Students will learn the **philosophy** and practical applications of design.

Students will become familiar with both **top-down designs** and **bottom-up designs**. This involves the advantages and challenges associated with each type of design. Differentiating main user functions and base functions is a major component of this section.

In hands-on workshops, students will learn how effective designs are created using each method. The course also covers problem solving in software design. This section will focus on **functional decomposition**. Students will learn to **decompose** software down to its subfunctions. As a final project, students will create **primitive** software with **rational** design plans. This course is a prerequisite for many advanced software courses. It prepares students for higher-level concepts like **stepwise refinement**.

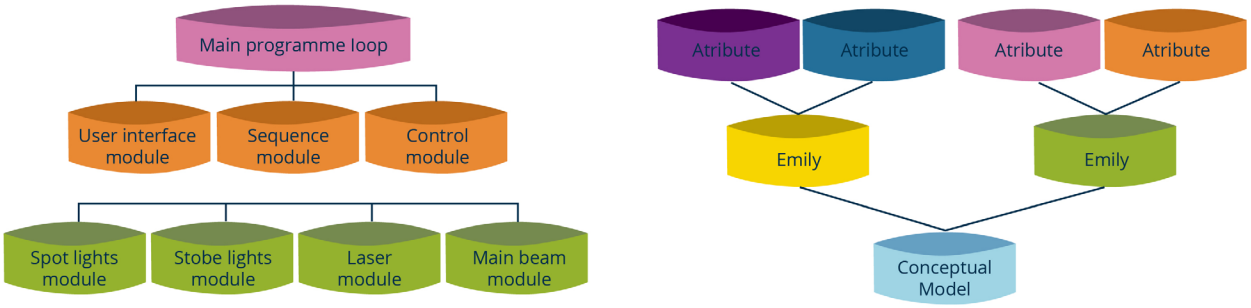



*Fig.1. Introduction to design methods.*

*Audio Reading. "Introduction to design methods"*
*https://on.soundcloud.com/KhsjGZu5gRQqPokD7*

**1. Read the course description. Then, choose the correct answers.**

**A. What is true of the design course?**

a. It requires a prerequisite.

b. It focuses on the history of different design methods.

c. It teaches advanced-level computing concepts.

d. It covers the philosophy of software design.

**B. Which of the following is NOT a topic covered in the course?**

a. top-down designs

b. stepwise refinement

c. decomposing software

d. functional decomposition

**C. What will students do for their final projects?**

a. design very basic software functions

b. make a diagram of bottom-up designs

c. write about the philosophy of design methods

d. make a plan for problem solving

**2. Read the sentence pairs. Choose which word or phrase best fits each blank.**

*1.1. top-down design / bottom-up design*

A. A _____ begins with the main user functions.

B. A _____ begins with the most primitive functions.

*2.2. primitive / idealistic*

A. A(n)_____ person ignores possible problems that may arise.

B. If a software is_____ , it is very simple.

*3.3. philosophy / subfunction*

A. A _____ is combined with others to make a function.

B. _____ provides a means of viewing the world.

# / 4. Soluciones

## Solución ejercicio Listening

**1. Listen and complete the conversation.**

1) managing

2) eliminate

3) top

4) easier

5) decomposing

## Solución ejercicio Reading

**1. Read the course description. Then, choose the correct answers.**

**A. What is true of the design course?**

a. It requires a prerequisite.

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c. It teaches advanced-level computing concepts.

***d. It covers the philosophy of software design.***

**B. Which of the following is NOT a topic covered in the course?**

a. top-down designs

***b. stepwise refinement***

c. decomposing software

d. functional decomposition

**C. What will students do for their final projects?**

***a. design very basic software functions***

b. make a diagram of bottom-up designs

c. write about the philosophy of design methods

d. make a plan for problem solving

**2. Read the sentence pairs. Choose which word or phrase best fits each blank.**

*1.1. top-down design / bottom-up design*

A. A **top-down design** begins with the main user functions.

B. A **bottom-up design** begins with the most primitive functions.

*2.2. primitive / idealistic*

A. A(n) **idealistic** person ignores possible problems that may arise.

B. If a software is **primitive**, it is very simple.

*3.3. philosophy / subfunction*

A. A **subfunction** is combined with others to make a function.

**B. Philosophy** provides a means of viewing the world.